

Tsinghua University graduate Student examination

Examination Course: Digital VLSI Design (FINALTERM) 2021.12.30

(120 minutes)

I. Choice question【26 points in total, 3 points for each single-choice question and 4 points for each multiple-choice question (no score for wrong selection, 1 points for missed selection)】

1) There is a logic path composed of 5 stages of identical inverters. The sizes of the transistors in each stage of inverter are: $(W/L)_n=1/1$, $(W/L)_p=2/1$. It is known that the random variation ($\sigma I/I$) of the on-current of a MOS transistor with $W/L=1/1$ caused by process variation is 5%, and hence, as a result, the corresponding variation of the total delay of the path ($\sigma D/D$) is ____ (single choice)

- A. $>5\%$ B. $=5\%$ C. Between $(\sqrt{5}\%, 5\%)$ D. $=\sqrt{5}\%$ E. $< \sqrt{5}\%$

2) The right statements about Network-on-chip in the following are ____ . (multiple-choice)

A. NoC has higher scalability than bus interconnection structure, i.e., its performance of communication is more insensitive to the number of IPs.

B. NoC can conquer the long delay problem in long-distance interconnections

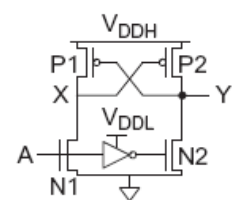
C. NoC should be widely used in all kinds of digital VLSIs to replace the traditional bus style fabric.

D. Nodes in a NoC communicate each other in parallel, hence NoC has higher communication bandwidth

3) Which of the issues below should be addressed for power and ground networks during digital VLSI layout routing? (multiple-choice)

- A. $L di/dt$ B. IR drop C. electromigration D. cross-talk

4) The right figure shows a receiver of the single-end low-swing transmission system, its input, A , swings from 0 to V_{DDL} and the supply voltages $V_{DDL} < V_{DDH}$. Then, this circuit is a ____ logic circuit. (single choice)



A. ratioed

B. ratioless

5) In a bus interconnection system, the propagation delay between units is a ____ function of the number of units connected to the bus. (single choice)

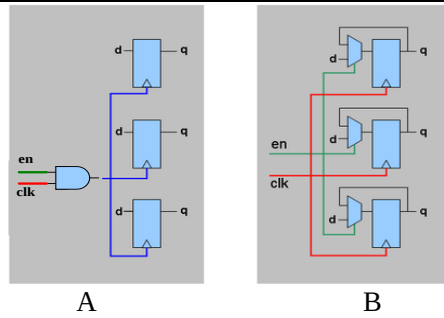
A. linear

B. logarithmic

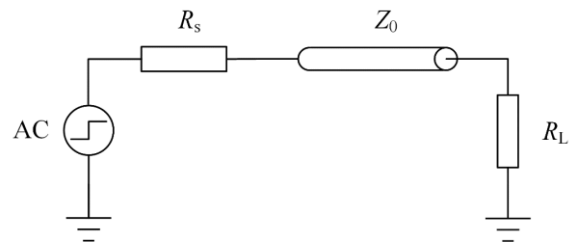
C. quadratic

D. exponential

6) Among the two following structures of register with enable-signal, ____ is more beneficial to the clock quality. (clk is the clock signal, en is the enable signal) (single choice)



7) As shown in the right figure, the signal source with the driving resistance R_s drives a transmission line with characteristic impedance equal to Z_0 , and the terminal at the receiving end is open ($R_L = +\infty$). In which of the following situations, the signal propagation delay on the transmission line is the shortest? ____ (single choice)



- A. $R_s = Z_0/2$ B. $R_s = Z_0$ C. $R_s = 5Z_0$ D. $R_s = 2Z_0$

8) When routing clock signals, we often increase the space between the clock signal and the adjacent wires. The primary reason for this is _____. (single choice)

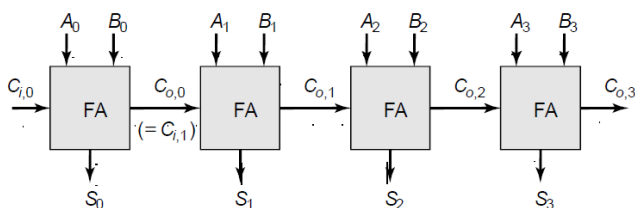
- A. To reduce the load of clock so that increase the signal flip speed.
- B. To provide more routing resources for the clock signal, shorten the clock signal wires.
- C. To reduce the crosstalk of the clock signal on the adjacent wires
- D. To reduce the crosstalk of the surrounding data signal wires to the clock signal so that reduce the clock jitter caused by the random switching signals.

II. True or false question [21 points, 3 points for each question]

1. The signal delay over a long wire is a quadratic function of its length.
2. The main disadvantage of the grid style clock distribution is large power dissipation.
3. Applying Delay-Lock-Loop in clock distribution can effectively suppress the clock jitter introduced by IR drop.
4. Low impedance nodes in circuits are more sensitive to the cross-talk between wires.
5. The noise margin of dynamic logic gate is the threshold voltage of a transistor.
6. There are not only a highest clock frequency permitted but also a lowest clock frequency permitted for a domino logic circuit.
7. DCVSL and Differential Domino Logic are both differential logics, the two output signals of a gate in both of them switch at the same time during transition.

III. Fill in the blanks 【16 points in total, 4 points for each blank】 :

1). There is a 4-bit ripple carry adder in the right figure. If the setup signals generation time of a full-adder (FA) is a , the carry path delay (from setup signals or C_i to C_o) is b , and the time of calculation sum (from carry-in(C_i) to S) is c , then the critical path delay of the whole adder is (1) ;



If this structure is expanded to an 8-bit adder, the critical path of the 8-bit adder will be (2) .

2. There is an inverter driving a load capacitance ($C_L=4\text{fF}$) through a 50um-long wire. The unit-length resistance and capacitance of the wire are $0.1\Omega/\text{um}$ and $0.06\text{ fF}/\text{um}$ respectively. The driving resistance of the inverter is 10Ω and the intrinsic load capacitance of it is 0.5fF . Then, according to the distributed rc model, the total delay, including the wire and the logic gate, is (3) ps.

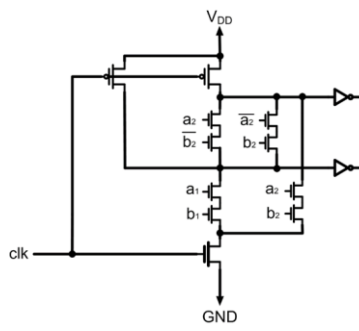
3. If a logic path is composed of N stages of pass transistors in cascade without inverters,, the relationship between the propagation delay of the pass transistor logic and N is a (4)th function.

IV. Subjective question

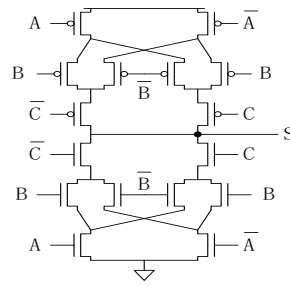
1. 【9 points】 Please write three failure mechanisms that cause wear and tear as the chip's use time increases. (Just list the name.)

2. 【12 points】 For a timing path composed of two identical rising edge triggered registers and a block of combinational logic, the setup time of both two registers is 0.8 ns , the propagation delay and contamination delay from the clock to output (clk-to-Q) are 0.6 ns and 0.5 ns respectively, the hold time of both two registers is 0 ns , and the combinational logic's propagation delay is 2 ns . When the clock skew is positive and equals to 0.5 ns , and the absolute jitter of each clock is 0.02ns , if the target clock period is 3 ns , please give the calculation process to show whether this timing path satisfies all timing constraints.

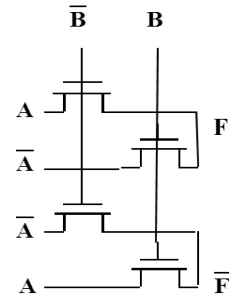
3. 【16 points】 Please give the function expressions of the gates below 【4 outputs in total, 4 points for each one】



(a) O1=?O2=?
(*clk* is a clock signal)



(b) S=?



(d) F=?